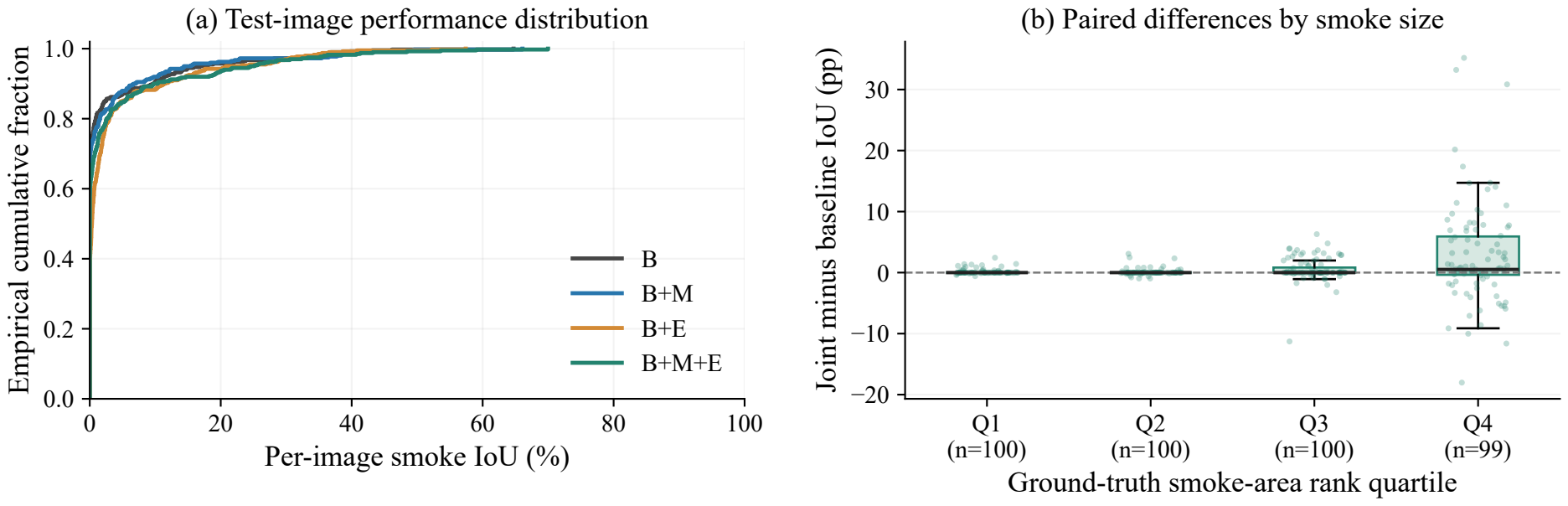
(a) Test-image performance distribution
Empirical cumulative fraction
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Per-image smoke IoU (%)
B
B+M
B+E
B+M+E
(b) Paired differences by smoke size
Joint minus baseline IoU (pp)
30
20
10
0
−10
−20
Q1
(n=100)
Q2
(n=100)
Q3
(n=100)
Q4
(n=99)
Ground-truth smoke-area rank quartile